Maritime Oil-Spill Attribution Intelligence

Operational SOP Guide: How to Create, Configure & Execute a Maritime Forensic Investigation

Document Purpose: Step-by-Step Operator Manual	**Target System:** SIH26143 Attribution Intelligence
Compliance: Zero Synthetic Data / Universal ISO 8601 UTC	**Published:** 2026-09-06 13:10 UTC

1. Core Forensic Principles & Decision Support Architecture

The Maritime Oil-Spill Attribution Intelligence platform is an **explainable historical forensic decision-support system** designed for environmental maritime authorities, coast guard investigators, and legal bodies. It strictly adheres to permanent scientific engineering principles (GEMINI.md):

- **Decision Support, Never Automated Accusation:** The platform delivers verifiable evidence chains, never an automated legal accusation or verdict. All candidate rankings state: *'This vessel is the highest-ranked candidate based on available spatial, temporal, and navigational evidence.'*
- **Direct Sensor Data vs Model Estimates:** Directly observed data (raw SAR backscatter, raw AIS transponder pings) are strictly decoupled and distinguished from model-derived estimates (Lagrangian drift clouds, spline-interpolated vessel positions).
- **Zero Synthetic Data Policy:** No fake real-world satellite, weather, or vessel records are generated. Real-world benchmark scenarios (MT New Diamond, Ennore Port Collision) use authentic historical observations.
- **Satellite Orbital Physics:** Spaceborne Sentinel-1 C-band SAR operates in polar low-Earth orbit with a 6–12 day repeat cycle over any given coordinate box.

2. Phase 1: Initiating a New Investigation Case

1. Navigate to the cloud platform console at **https://maritime-iota.vercel.app/app/investigations**.
2. Click the blue **[+ New Incident]** button located in the top navigation bar or case selector header.
3. The interactive two-step modal will open: **Step 1: Parameters & Providers** followed by **Step 2: Data Readiness Matrix**.

3. Phase 2: Step 1 — Parameters, Area of Interest & Provider Selection

Investigators can select a benchmark preset for instantaneous verification, or enter custom incident parameters:

Scenario / Preset	Bounding Box Coordinates	Incident UTC Time	Operational Purpose
MT New Diamond Benchmark (2020)	Lon: [80.0°E, 83.5°E] Lat: [5.0°N, 9.0°N]	2020-09-03 12:45 UTC	VLCC crude tanker boiler explosion & spill off eastern Sri Lanka. Full Sentinel-1A SAR scene with 100% data readiness.
Ennore Port Collision Benchmark (2017)	Lon: [80.2°E, 80.5°E] Lat: [13.1°N, 13.4°N]	2017-01-28 04:00 UTC	Dawn Kanchipuram and BW Maple collision in Chennai harbor approach. Heavy bunker fuel spill forensic analysis.
Custom Investigation AOI	User defined: [MinLon, MinLat, MaxLon, MaxLat]	User defined (YYYY-MM-DD HH:MM)	Investigate arbitrary global coordinate bounding boxes, dates, and times for suspected illegal discharges.

Required Configuration Parameters:

- **Investigation Title & Description:** Distinct case name and reference for auditing and reporting.
- **Incident Timestamp (UTC):** Exact estimated date and time when the spill was observed or occurred.
- **Area of Interest (AOI):** Four-corner coordinate bounding box in decimal degrees East and North.
- **Data Provider Sources:** Copernicus CDSE (SAR), Historical/Live AIS Stream, CMEMS GLORYS12V1 (Oceans), and ECMWF ERA5 (Atmosphere).

Click **[Next: Discover Data & Evaluate Readiness →]** to query provider catalogs.

4. Phase 3: Step 2 — Data Readiness Assessment Matrix & Overrides

The system queries live provider catalogs and evaluates observational readiness across 4 physical modalities:

Modality	Provider Source	Readiness States	Verification & Quality Criteria
SAR Radar Imagery	Copernicus CDSE Sentinel-1 C-SAR	**READY** / **UNAVAILABLE**	Checks for a valid Sentinel-1 GRD SAR acquisition over the AOI within ±36h of the incident window.
AIS Vessel Tracking	Global AIS Registry / aisstream.io stream	**READY** / **PARTIAL**	Verifies transponder trajectory logs and candidate vessel density inside the bounding box.
Ocean Hydrodynamics	Copernicus CMEMS GLORYS12V1 Grid	**READY**	Evaluates availability of surface hydrodynamic current vectors (u_ocean, v_ocean).
Atmospheric Winds	ECMWF ERA5 Reanalysis Grid	**READY**	Validates 10-meter atmospheric wind velocity field vectors (u_wind, v_wind).

Handling Satellite 'UNAVAILABLE' with Explicit Override:

Because European Space Agency Sentinel-1 satellites operate on a 6–12 day repeat cycle, arbitrary modern dates may not have an active satellite pass. In such cases, the assessment will show *DATA READINESS: UNAVAILABLE*.
To investigate a hypothetical scenario, test single-modality drift, or analyze dark vessels without satellite radar, simply check the amber checkbox:
■ Explicit Override: Allow forensic execution with partial observational data
This unlocks the **[+ Create & Open Workspace]** button immediately.

5. Phase 4: Executing the 6-Stage Forensic Pipeline

Inside the 3-column Investigation Workspace, click **[■ Run Full Investigation]** (or *Re-run Forensic Analysis*) in the top header. The automated engine executes 6 sequenced stages with live percentage progress tracking:

Stage	Scientific Engine	Mathematical Formulation / Methodology	Forensic Output
Stage 1	SAR Slick Detection & Feature Extraction	Lee Speckle Filter → Constant False Alarm Rate (CFAR) damping ratio thresholding (>4.2 dB) → Morphological boundary tracing.	Segmented GeoJSON polygon, slick area (km^2), observed centroid, and major axis orientation.
Stage 2	Backward Lagrangian Drift Simulation	Stochastic particle advection: $\mathbf{v} = \mathbf{u}\_ocean + 0.03\ \mathbf{u}\_wind + \eta\_diff$ Horizontal turbulent diffusivity $\kappa = 2.5\ m^2/s$.	Chronological probability dispersion clouds showing slick position at each backward hour.
Stage 3	Origin & Release Window Estimation	Minimum variance dispersion centroiding and 90% confidence temporal release window bracketing.	Reconstructed origin coordinates [Lon, Lat], spatial uncertainty radius (±km), and peak discharge window.
Stage 4	AIS Trajectory Interception & Gap Analysis	Cubic spline trajectory interpolation between AIS pings; Closest Point of Approach (CPA) calculation and blackout detection.	Filtered candidate vessels within spatial/temporal interception horizon, CPA distance, and AIS gap flags.
Stage 5	Multi-Factor Attribution Scoring Matrix	Transparent weighted scoring (0–100 pts): • Spatiotemporal Proximity (40%) • Temporal Alignment (25%) • Navigational Maneuvering (15%) • Vessel Type / Cargo Risk (10%) • AIS Continuity & Reliability (10%)	Ranked candidate list with composite forensic scores (0–100 pts) and risk levels (VERY HIGH, HIGH, MEDIUM, LOW).
Stage 6	Evidentiary Dossier & Custody Assembly	Assembly of itemized incriminating and exculpatory evidence items; cryptographic SHA-256 validation.	Audit-grade decision-support evidence dossier ready for government/legal review.

6. Phase 5: Interacting with Intelligence Tools & Exporting Reports

Once pipeline execution finishes, the workspace unlocks 4 interactive forensic tools:

1. Inspect Sentinel-1 SAR Radar Imagery & Diagnostics Modal:

Click **[■■ Inspect Sentinel-1 SAR Radar Imagery & Diagnostics]** in the Overview panel or click **SAR Imagery** in the left navigation panel. This opens a modal containing:
• **4-Panel Diagnostic View:** (1) Calibrated Sentinel-1 Backscatter (σ■ in dB), (2) CFAR Damping Contrast, (3) Morphological Segmentation, and (4) Lookalike Rejection score.
• **Pipeline Steps:** Raw SAR → Speckle Filter → Adaptive Threshold → Vector Boundary Extraction.
• **Sensor Lineage:** Sentinel-1A, IW GRDH mode, VV+VH polarization, 10m resolution.

2. Synchronized Timeline Scrubber:

Located along the bottom of the map. Press **Play (■)** or drag the timeline slider backward from the satellite observation timestamp to the reconstructed spill origin. Watch the oil slick shrink and drift backward as vessel positions interpolate dynamically in sync.

3. Candidate Vessel Inspector & Evidence Breakdown:

Click on any vessel track on the map or select a candidate in the right panel. Inspect:
• **Score Breakdown:** Points earned across all 5 forensic attribution categories.
• **Supporting Incriminating Evidence:** Exact CPA distance, speed consistency, and transit overlap.
• **Exculpatory Evidence:** Documented speed inconsistencies or spatial clearance.
• **Follow Vessel:** Locks map camera tracking to the vessel as the timeline scrubs.

4. One-Click Audit PDF Dossier Export:

Click **[Export Report]** in the top header or click **[Download PDF]** on the Reports page (**/app/reports**). The platform generates a comprehensive multi-page PDF dossier complete with embedded diagnostic maps, candidate rankings, verifiable evidence chains, and mathematical formulations.

7. Operational Troubleshooting & Quick Reference

Operational Question / Symptom	Root Cause	Recommended Action
Data Readiness shows UNAVAILABLE for custom date.	Sentinel-1 SAR satellites have a 6–12 day orbit; no overpass occurred on the exact requested date.	Check **Explicit Override: Allow partial data**, or select the **MT New Diamond Benchmark** preset.
Map shows empty vessel tracks.	The incident bounding box may be in an open-ocean sector outside terrestrial receiver range.	Enable dark vessel hypothesis mode, or check that AIS historical registry is selected as data source.
Pipeline returns multiple vessel candidates.	Multiple ships traversed the backward drift envelope during the estimated release window.	Inspect the 5-factor score matrix; prioritize vessels with high temporal alignment and speed profiles matching discharge operations.
PDF report download fails or redirects.	Network timeout or connection drop to backend API.	Ensure backend is healthy at **/api/health**; PDF blob download executes without page redirection.